

Urban Water Policy in the Eastern Himalayan State of Sikkim: An Assessment

LAKPA DOMA SHERPA

Assistant Professor, Department of Economics, Government College, Rhenock, East Sikkim, Sikkim

KOMOL SINGHA

Associate Professor, Department of Economics, Sikkim University, Gangtok, Sikkim

Abstract

The urbanisation process in the eastern Himalayan state of Sikkim has been very rapid in the recent past. As a result of this, the demand for safe drinking water in the urban centres has also increased rapidly. However, despite this rapid urbanisation, residents have so far been getting sufficient and safe drinking water. But the Government of Sikkim (GoS) is now increasingly finding it difficult to supply uninterrupted safe drinking water, especially in the East Sikkim district, with the nominal revenue generated from the present non-volumetric water supply system. The GoS has therefore proposed a new water policy, a volumetric/meter system though the residents are quite reluctant to switch over to the new policy in which they are required to pay a slightly higher tariff than what they have been paying so far. However, under the new policy, the system seems to be better positioned in terms of making the water users accountable and the service provider economically sustainable.

Keywords: Eastern Himalaya, Sikkim, Urbanisation, Water Demand, Water Pricing Policy

Background

The urban water resources management, especially in trying to maintain a balance in the environment and make water supply sustainable has become an important issue due to the rapid urbanisation in the recent past (Cosier & Shen, 2009; Gupta, 2001). In most countries, government agencies act as a regulator and distributor of water service delivery to the public. Water being merit good, it is provided mainly by the government at a subsidised tariff (Cosier & Shen, 2009). As a result of this, over the years its supply has been under-priced for the public good (Aggarwal, Maurya & Jain, 2013; Hirshleifer & Milliman, 1967), primarily due to failure in its rational reallocation (Hazra & Barua, 2020). The allocation of water resources among different sectors- domestic consumption, industries, agricultural use etc.- has been increasingly facing a major challenge nationally as well as internationally (Kumar & Pandit, 2016; Gupta, 2001). Cities are rapidly expanding and therefore require more and more water for human use and for commercial activities, and all urban centres require higher per capita water compared to the rural areas (Kumar & Pandit, 2016). As the state has been providing water to the public at subsidised rates, it is increasingly constrained by financial hardship and this has led to a lower quality of water supply, insufficient infrastructure and other unforeseen maintenance expenses

(Aggarwal, Maurya & Jain, 2013). In the case of the Australian water crisis, Sibly (2006) pointed out that the Sydney Water Board's volumetric rate was systematically underpriced. Therefore, in 1988-89, the Sydney Water Board adopted a new format called Increasing Block Tariff (IBT), in which the water provider sets a number of tiers of consumption level, and a low volumetric rate is set in the lowest consumption tier (tier 1) with the rate increasing in subsequent higher tiers.

The urbanisation trend in India's north-east¹ (or region hereafter) was found to be quite asymmetric, concentrated mainly in the state capital cities. According to Singh and Singha (2020), the "reason for unfrequented development is quite clear, and this is not surprising, given the topography, geopolitical environment, population density and geographical isolation from the rest of the country." As a result, the demand for drinking water in these cities has increased rapidly, and at the same time, the demand and supply water gap has also increasingly widened in the recent past. For instance, only around half the households in Guwahati Metropolitan City could access piped water (Hazra & Barua, 2020). In Manipur, Chief Minister of Sikkim was compelled to convene an emergency meeting on 23 March, 2021 to review the acute water crisis in the capital city, Imphal. Here, the supply of tap water has been greatly affected due to the drying up of the Singda Dam, the main water reservoir which supplies drinking water to Imphal city (Leivon, 2021). In a recent study by Jain, et al. (2021), the groundwater in Shillong city, the capital of Meghalaya, was badly affected by ever-increasing industrial activities. In Agartala, the capital city of Tripura, the quality of drinking water was found to be deteriorating rapidly due to the rapid increase in population and poor maintenance of the water pipelines. So much so that in 2018, a large number of people from Agartala were hospitalised due to the consumption of contaminated drinking water supplied by the municipality (Deb, 2018).

In the Himalayan state of Sikkim², the urbanisation trend seems to be quite skewed, concentrated in the capital city Gangtok and its surrounding towns, which fall under East Sikkim district. For instance, around 70 per cent of the state's registered vehicles were found in and around Gangtok city (Singha & Raj, 2019). With the rise in population, urbanisation and industrialisation, together with the spatial and temporal variations in water supply, the need for an active and effective policy on water resources management in this hill city has been realised in the recent past. According to a study conducted by Tiwari (2012) it was found that the South Sikkim district—the second most populated district in Sikkim—suffers not only from the shortage of water supply but also from ignorance of the quality of available water as well as water conservation techniques. In a recent study conducted by Sharma, et al. (2020) it was found that the demand for drinking water in Singtam town—one of the fastest growing towns in East Sikkim district—was increasing rapidly due to increase in the floating population, and at the same time, the water sources, especially the natural springs, were also found to be drying up at an alarming rate. The situation was further compounded by the inappropriate water reservoir infrastructure and the frequent damage of water supply pipes during the monsoon. Interestingly, the same study further highlighted that the water shortage was not faced equitably by all residents as there are alternative water sources. It is the migrants and the floating population who are mainly

¹North-east India consists of eight states—Arunachal Pradesh, Assam, Manipur, Mizoram, Meghalaya, Nagaland, Sikkim and Tripura.

²Sikkim, the erstwhile independent Buddhist kingdom—located in the eastern Himalaya—merged with the Indian Union on 16 May, 1975, became the 22nd and the second smallest state of India. This landlocked hilly state covers an area of 7096 sq. km. with 6.11 lakh population (Census of India, 2011), is divided into four districts as—East, West, South and North. Gangtok, the capital city, fall under East Sikkim district.

facing a water problem. Sherpa (2019) however found that despite Sikkim's rapid urbanisation and population growth in the recent past, the issue of sufficient domestic water supply has not been very serious in the East Sikkim district, especially in the urban centres. The Government of Sikkim (GoS) has been able to supply adequate and safe drinking water to the urban centres. There is also an alternative water supply system in the state, along with private line connections which are purely managed by private individuals.

With this background, the present study tries to answer a few research questions: 1) what makes the water supply regular and uninterrupted in East Sikkim when other cities within the region face an acute water shortage; 2) what are the government water policies in place in East Sikkim district in providing uninterrupted water supply; and 3) whether the residents are happy with the state water policy or not.

Objectives and Methodology

The basic concept of water utility has been well defined, in terms of per capita water availability for consumption and the hours for which water is supplied per day (Howard & Bartram, 2003). The service delivery of water in Indian cities over the years is continuously inadequate and the quality was found to be unreliable. The sector suffers from chronic operational inefficiency due to the lack of infrastructure development and expenditure on it. The problem is further aggravated due to an uneconomic tariff structure, low collection efficiency and high level of leakage of water from pipes lying on the surface of the earth (Tiwari & Gulati, 2011). However, the problem is still not so acute in the Himalayan state of Sikkim. With this background, the present paper tries to investigate the existing urban water policy, especially the water supply/distribution system, in the eastern Himalayan state of Sikkim. Whether the people in urban centres, especially in the East Sikkim district are given sufficient water or not is discussed. The paper further tries to assess the people's opinion on the existing water policy of the state. Should the public be able to choose any alternative water policy is also discussed.

To achieve the above-mentioned objectives, both primary and secondary data have been employed. With the help of a well-structured questionnaire, primary data were gathered from 355 urban households (280 samples from Gangtok, 25 from Singtam, 30 from Rangpo and 20 from Rhenock) under the East Sikkim district. We have used multi-stage sampling technique to identify the final household or respondents. In the first stage, we had identified the district having the highest percentage of population in Sikkim. Based on the criteria, defined by the Population Census of India 2011, we selected three urban centres and one census town from the East Sikkim district. From the selected urban centres, we further narrowed down to the wards of each urban centre. Since Gangtok is the most populated city followed by Rangpo, Singtam and Rhenok, we proportionately selected the sample households from the above mentioned urban centres to reach our total sample of 355. The primary field survey was conducted during June to August 2018. The gathered data basically include socio-economic factors that might have influenced water consumption, demand and supply of domestic water, sources of water, preference of supply policy etc. Also, by giving two options- 1) existing non-volumetric water policy, and 2) the government's proposed new meter (volumetric) water supply system—the respondents were asked their preference of a water supply system in the urban centres of East Sikkim district. This exercise provides us an understanding of the existing water policy and whether people are satisfied with it. As for the secondary data, information provided by the

Water Security and Public Health Engineering Department, Government of Sikkim, was gathered and descriptive statistics have been employed to analyse the state water policy.

The management of domestic water resources in Sikkim is basically divided into two types—public water supply managed by the state government, and private water supply managed by private actors. In the former category falls the water supply provided by the government through different channels of supply, and maintenance of water supply networks by a government agency, i.e. the Public Health Engineering Department (PHED). Private water supply deals with the various water extraction sources on the surface, such as glaciers, rivers and springs by individuals/community. The government is not involved in this case. No regular tariff is paid for this kind of water pipe connection, but the expenses incurred in making a reservoir at the source, laying a pipeline and its repairs, cleaning the reservoir from time to time etc. are shared by the users. Very negligible residents in the East Sikkim district were found to be connected with the private pipeline as their main source of water supply. For instance, in our field survey, 50 out of 355 sample households (14 per cent) were found to be connected to the private pipeline as their main source of water supply, and they were mainly from the periphery of the city. The private pipeline is basically connected as a standby/back-up system to the public (PHED in this study) pipeline.

The main water supply source for the Gangtok Municipal Council area is Ratey-chu, which emerges from the Himalayan glacier-fed lake at Tamze. Drinking water is tapped at an elevation of 2,500 meters, transported for around 17 kilometres of pipelines to the Selep Water Treatment Plant, Gangtok, and distributed to 19 wards. For the Nagar Panchayat areas of Singtam, the main water source is located at Chalamthang, which is 11 kilometres from Singtam. For the Rangpo, the main water source is located at Kamarey and Dikling, which are around 26 kilometres from the town. For the Rhenock town, the main water source is located at Sudunglakhang and Kopchi, which are around 5 kilometres from the destination.

Water Policy in Urban East Sikkim District

Water being a state subject under the Constitution of India, the government has the primary role and responsibility to use and control the water sector. Since the 1950s, various schemes and programmes have been implemented regarding drinking water supply in India. The first National Water Supply and Sanitation Programme was launched in 1954 for providing safe drinking water to urban and rural areas and the first National Water Policy came into existence in 1987 under the Ministry of Water Resources, Government of India. In the Indian context, providing adequate and safe drinking water has been a challenge, unlike European countries where adequate quantities of water with appropriate quality are provided (Pandit & Biswas, 2019).

In the case of Sikkim, in 2007, the State Water Policy was framed in line with the National Water Policy of 1987 for water resources management and development in the state. With the expansion of the economy, there has been a drastic increase in demand for water in various sectors, namely, domestic, hydro-power, industries, agriculture and recreation activities.

Sikkim is known for its substantial water resources and good quality due to its rich glacier-fed rivers, streams and springs. At present, the state has around 1.05 lakh households, of which 70,525 (67 per

cent) have got tap water connections. The state plans to provide 100 per cent tap water connections to all its residents by the year 2021-22. For enabling this mission, the central government has allocated Rs 31.36 crore to Sikkim for the implementation of the Jal Jeevan Mission in 2020-21, of which Rs 7.84 crore has been released. Another Rs 42 crore has been allocated under the 15th Finance Commission Grants for rural local bodies of the total allocated fund, 50 per cent of it to be used for watersupply and sanitation (The Sentinel, 22 October, 2020). For the 2021-22 budget allocation, a Rs 50,011 crore budget allocation has been made for the same mission. Further, Rs 26,940 crores are available as an assured fund under the 15th Finance Commission tied-grants to RLBs/PRIs for water and sanitation, with a matching state share and externally aided projects. In this manner, in 2021-22, the state has planned more than Rs 1 trillion for ensuring tap water supply to rural homes. With this enhanced budgetary allocation, Sikkim is likely to get almost 2.5 times more central funds in 2021-22 as it did in 2020-21 (Mint, 2021).

On the other hand, in the state the rainfall rate was declining over the period 2011 to 2014 (Sherpa, 2019). The availability of water from rainfall showed an inconsistent pattern. There were also reports of natural springs drying up in the state (Sharma et al., 2020). The state's population is unevenly distributed, wherein the east district of Sikkim is the most densely populated, but the district recorded the largest number of families with access to tap/piped water at 89.9 per cent of households (Census of India, 2011). A similar figure was also supported by our primary field survey conducted in the month of June-July 2018, estimated at 83.66 per cent of households with access to tap/piped water. According to the Sikkim Water Supply and Water Tax Act 1986, the state government takes responsibility for the construction and maintenance of waterworks. While providing water supply, the first priority is given to domestic consumption, followed by commercial and then other sectors subject to the availability of water (GoS, 1986). Further, the existing rules and regulations of the Sikkim Water Supply Act 1990 regulated by the PHED mentions that the distribution network of water supply was carried by a non-volumetric approach. This was done based on the number of taps used for domestic and commercial purposes. Details are given in Table 1.

Table 1: Present Non-Volumetric Water Tariff Policy in Sikkim

No. of Tap Connections	(in Rs./Month/Tap)			
	Domestic	Change with Slab/Tier Rises (%)	Commercial	Change with Slab/Tier Rises (%)
1-3	90	—	180	—
4-7	150	66.7	290	61.1
8-12	250	66.7	390	34.4
Every tap above 12	50	—	100	—

Source: Authors' Estimation from GoS (2014), GoS (1986)

Table 1 depicts the water tariff based on the non-volumetric distribution in rupees per month per tap connection in Sikkim. GoS made further rules to amend the Act of 1990 called the Sikkim Water Supply Rules 2014. Under this amendment, water supply services would be rationalised with the introduction of a volumetric tariff that would replace the existing tap-based tariff. From Table 1 we can see that as the number of connections increases from the range of 1-3 taps to 4-7 taps, the tariff increases from

Rs 90 to Rs 150 per month for domestic connections and Rs 180 to Rs 290 per month for commercial connections. Further, as the number of tap connections increases from 4-7 to 8-12, the tariff also rises from Rs 150 to Rs 250 for domestic and Rs 290 to Rs 390 for commercial connections respectively. But when the number of tap connections further increases above 12, the tariff for every increase in tap will attract Rs 50 for domestic and Rs 100 for a commercial connection. For example, a family having 13 water taps will be required to pay Rs 300 (Rs 250 plus Rs 50) for a domestic connection. The major issue in this policy is the rate of change in tariff as the slab/tier of tap connections rises. In the case of domestic connections, the change in tariff with the increase in the slab/tier of a tap connection remains the same (66.7 per cent) from the 1-3 to 8-12 slab of tap connection, and Rs 50 for every tap above 12 connections. For commercial connections, the rate declines as the slab of tap connections rises—61 per cent to 34 per cent when it rises from 4-7 to 8-12 tap connections respectively. Under this present water policy, it is seen that the tariff is very low, and is not an incremental change in tariff as the slab/tier rises. It is a regressive pricing policy.

The ever-increasing population and its corresponding demand for water in East Sikkim district, especially in and around Gangtok areas has been realised by the GoS. According to the recommendation given by the Ministry of Urban Development (MoUD), the water supply requirement for the Gangtok area can be met by a metering system of water supply and replacing the smaller diameter pipeline called the Secondary Feeder Mains (SAAP, 2016). Based on this amendment and by understanding the drawbacks of the present non-volumetric/tap-based tariff, the Water Supply Rules 2014, the GoS has given the volumetric tariff-based rate for water users in Sikkim. Of course, it has not been operational to date. The detail is given in Table 2.

Table 2: Proposed Volumetric Water Tariff in Sikkim

Volume	(Rs./Month/1000 Litres)			
	Domestic	Change with Slab/ Tier Rises (%)	Commercial	Change with Slab/ Tier Rises (%)
First 20 KL	5	-	10	-
20-40 KL	8	60.0	13	30.0
40-60 KL	12	50.0	17	30.8
Every KL above 60 KL	20	-	20	-

Source: Authors' Estimation with the inputs from GoS (2014)

From Table 2 it is found that the tariff rises from Rs 5 to Rs 8 and Rs 12 per 1,000 litres (KL) for the domestic connection and for the commercial connection, the rate rises from Rs 10 to Rs 13 and Rs 17 per 1,000 litres (KL), when the volume of slab/tier rises from first 20 KL to 20-40 KL and 40-60 KL respectively. But when the volume connection slab further increases above 60 KL, the tariff for every increase per 1,000 litres (KL) will attract Rs 20 for both domestic and commercial connections. For example, if a family consumes 61 KL of water they will be required to pay Rs 32 (Rs 12 plus Rs 20).

When compared to other major cities in the region, the tariff proposed by GoS is relatively still lower. For instance, the domestic water connection tariff at the Guwahati Metropolitan City at present is given as— Rs 8 per KL for the quantity up to 15 KL, Rs 12 per KL between 15-25 KL and Rs 15 per KL

for the consumption above 25 KL (Hazra & Barua, 2020, 166). Nevertheless, this proposed volumetric tariff on water supply in Sikkim will check the wastage level by residents and make them accountable for water leakage caused by pipe line damage or pilferage. Whether the public at large is happy about the proposed water policy in Sikkim may be evaluated scientifically in the following section.

Resource Allocation for Water Supply in Sikkim

Water supply for the urban centres, especially Gangtok city in East Sikkim district is covered by a central programme, the Jawaharlal Nehru National Urban Renewal Mission (JNNURM) along with Atal Mission for Rejuvenation and Urban Transformation (AMRUT). The remaining urban centres of the district provide water supply with funding from the various programmes implemented by the central as well as the state government. The growth rate of the annual state plan outlay/expenditure and revenue for water supply in Sikkim is depicted in Table 3.

Table 3: Plan Outlay & Revenue Collection from Water Supply in Sikkim

(Rs. in Lakhs)

Year	State Plan Outlay	Revenue Collection	Revenue Deficit
1	2	3	4=(2-3)
2002-03	350	73	277
2003-04	950	74	876
2004-05	875	95	780
2005-06	855	101	754
2006-07	961	197	764
2007-08	1755	202	1553
2008-09	1937	252	1685
2009-10	2007	253	1753
2010-11	1831	254	1576
2011-12	1541	288	1253
2012-13	3039	272	2767
CAGR	17	17	17

Note: CAGR = Compound Annual Growth Rate

Source: Authors' estimation from GoS (2014); GoS (2018)

The year-to-year growth rate of the state plan outlay and revenue collection was 171.43 per cent and 1.95 per cent respectively in 2003-04. On an average, the growth rate of the state plan outlay was 34.28 per cent and revenue collection was 16.57 per cent over the period 2003 to 2012. These figures indicate that the state plan outlay is double that of the revenue collected from water users in Sikkim from 2003 to 2012. The revenue deficit is given in the last column (column 4) of Table 3 and it is seen that the deficit level has been steadily increasing. This might have been one of the major concerns for the state to plan for a volumetric water policy.

Table 4: Expenditure on Water Supply and Sanitation in Sikkim

Year	Expenditure on Water Supply and Sanitation (Rs. in Lakhs)	Total State Expenditure (Rs. in Lakhs)	Share (%) of Expenditure on Water Supply and Sanitation to Total State Expenditure	Per Capita Expenditure on Water Supply and Sanitation (in Rs)
1985-86	116	7758	2	33
1986-87	152	8126	2	42
1987-88	773	9846	8	210
1988-89	229	11316	2	61
1989-90	227	11527	2	59
1990-91	291	12815	2	73
1991-92	374	15509	2	92
1992-93	412	17973	2	99
1993-94	409	18891	2	95
1994-95	531	52642	1	120
1995-96	651	88119	1	143
1996-97	701	111886	1	150
1997-98	715	125810	1	148
1998-99	964	149560	1	194
1999-00	1089	150997	1	213
2000-01	1020	76331	1	194
2001-02	1046	166425	1	193
2002-03	1161	188262	1	212
2003-04	1288	118093	1	232
2004-05	1224	172349	1	218
2005-06	1400	176761	1	247
2006-07	1498	188756	1	261
2007-08	1775	234856	1	305
2008-09	2360	238075	1	401
2009-10	2073	256715	1	348
2010-11	2484	290753	1	412
2011-12	2219	355904	1	363
2012-13	1651	357002	0	267
2013-14	2219	345796	1	355
2014-15 (BE)	2837	373095	1	448
2015-16 (RE)	4149	431275	1	647
CAGR	10	15	-5	8

Note: India. RE = Revised Estimate, BE = Budget Estimate

Source: Sherpa (2019)

The water supply and sanitation expenditure of the GoS is summarised in Table 4. The percentage of expenditure on water supply and sanitation to total expenditure in Sikkim was found to be less than 1 per cent during the period 2004 to 2015, while the state's population was increasing every year to the tune of 1.22 per cent (Census of India, 2011). The per capita water user charge showed a consistently increasing trend from Rs 33 to Rs 647 from 1985-86 to 2015-16.

Table 5: Maintenance and Operation Cost on Water & Sanitation in East Sikkim

Year	Plan (Rs in Lakhs)	Non-Plan (Rs in Lakhs)	Total (Rs in Lakhs)
2004-05	49	152	201
2005-06	73	137	209
2006-07	124	123	247
2007-08	140	135	275
2008-09	214	145	359
2009-10	63	147	210
2010-11	200	169	369
2011-12	236	154	390
2012-13	183	170	352
2013-14	182	163	345
2014-15	182	170	353
2015-16	130	296	426
CAGR	9	5	6

Source: GoS (1986); GoS (2014); GoS (2018)

Table 5 depicts the operation and maintenance expenditure incurred on water and sanitation in East Sikkim district alone from 2004-05 to 2015-16. The cost incurred in the district has been increasing constantly at 6 per cent per year. In absolute terms, it has been double the cost from 2004-05 to 2015-16.

In the rest of India's major cities also, an appropriate price setting reform for domestic water users is needed (Hazra & Barua, 2020; Sauri, 2013). The tariffs charged from urban water users are not only lower than the cost incurred for the same service (Mathur & Thankur, 2003), but also the revenue generated from the tariff has been steadily decreasing. The tariff for water usage charged in India is 20 to 25 per cent lower than that of the operation and maintenance costs (Mathur & Thankur, 2003). The situation is more serious in Sikkim in this regard. From the above descriptive statistics, we can summarise that the Sikkim government's deficit in water supply (cost-revenue gap) has been rising. Therefore, an appropriate price/tariff restructuring is very necessary. If an appropriate pricing mechanism is made, it could lead to the proper management of resources allocation not only for revenue generation but also for proper utilisation of the available water resources.

According to Sauri (2013), there is evidence of a negative relationship between the higher water pricing and water wastage and pilferage. For efficient management of water supply to the public, the government should collect at least the maintenance and operation expenditure on water supply. The revenue collected from the water users in Sikkim is unusually low compared the expenditure. This led to difficulty for the GoS to recover the maintenance and operation cost of water supply in the East Sikkim district. Of course, the situation is not different in other Indian cities. For instance, the Government of National Capital Territory (NCT) of Delhi is providing 20,000 litres of water per month free of charge to around 5.3 lakh households, and consumption above this will invite charges in Delhi. But this model cannot be applied in Sikkim. Therefore, GoS is planning to introduce the new volumetric policy mentioned above in Table 2.

Public Choice of Water Supply System in East Sikkim

From the above analysis, it is clearly seen that the state of Sikkim needs an appropriate urban water policy, especially concerning the pricing mechanism. To assess peoples' acceptance of the proposed water policy, in this paper we conducted a choice-based experiment for water service delivery in the month of June-July, 2018 and the respondents were asked to select one of the options which they consider the best out of the two options given to them.

Table 6: Choice-based Experiment for Water Service Delivery in East Sikkim

Option/Choice	1	2
Attribute	Existing System	Proposed by the Government
Connection	Non-meter (Non-volumetric)	Meter (Volumetric)
Supply/Quantity	Limited supply (approx. 1-3 hrs/day)	Regular (seemingly)*
Governance	Govt.	Govt.
Rent/Bill rate	Normal/low (Per connection)	Normal/moderate (Per unit)

Note: *As per the norm, water supply should be 24x7, but this does not happen in most of the cities in India

Source: Authors' compilation

Table 6 shows two options—the existing non-volumetric (tap-based) and the new/proposed meter (volumetric) water supply system. Option 1 represents the (non-volumetric) water supply system that is presently in place in Sikkim, and option 2 represents the new (proposed) water policy (volumetric). the tariff and detail slab/tiers are given in Table 1 and Table 2. The public response to the new policy from the field survey is given in Table 7.

Table 7: Choice-base Experiment of Water Policy in Urban East Sikkim District

Choice/Option	Number of Households	Percentage of Households
Existing non-volumetric system (Choice 1)	180	50.7
New/proposed volumetric system (Choice 2)	175	49.3
Total	355	100

Source: Primary Survey (June- August 2018)

From Table 7 we see that around 51 per cent of households still prefer the existing system of water supply (non-volumetric/scenario 1), and 49 per cent of the households prefer the meter system (volumetric) proposed by the GoS. From this exercise, we can understand that most of the households in urban East Sikkim district are still happy with the existing water supply system (without meter/option 1) for domestic consumption. This further endorses the service delivery of the present water supply in Sikkim, especially in the East Sikkim district as being satisfactory. No individual/resident would like to spend more money when the present policy benefits them, in which they are required to pay very little and still get a sufficient quantity of water for domestic consumption. This also indicates that the domestic water supply in East Sikkim district is not seen as a problem despite the rapid urbanisation taking place in the state. However, the GoS is now experiencing financial difficulties in providing sufficient and safe drinking water at the existing rates. Because of this the quality of water supply has been seemingly sacrificed, albeit this section is not covered in the present study.

Summary and Conclusion

As regards the supply of water for domestic consumption, the residents of East Sikkim district are getting sufficient quantities as they are backed by e private pipelines. This has been made possible due to the state's abundantly available water sources, including the springs and glacier-fed streams. For instance, according to a government report, around 40 seasonal local springs are found in the East Sikkim district. Though the exact number has not been estimated by this study, a fairly large number of households in Gangtok city are using local springs, such as the Nam-Nang fall/spring below Mahatma Gandhi (MG) Marg, Gangtok. However, for commercial activities, especially in Gangtok city, when there is any water crisis due to the breakage of water pipelines, landslides, repairing water pipelines etc., private water tankers have to be used which fetch water from the nearby springs (e.g. Nam-Nang spring and 2nd Mile spring at Chandmari JN Road). On the other hand, the GoS/PHED seems to be struggling in providing sufficient and safe drinking water to the ever-increasing urban population of the East Sikkim district. Looking at the low revenue collection from water users, one can only imagine the amount of financial burden having to be borne by the state. The expenditure on water and sanitation in Sikkim was found to be double the revenue generated from water users.

For the growth and development of water resources management in the urban centres, an appropriate water policy that includes infrastructure, pricing and institutional structure is necessary. Therefore, the new water policy (volumetric) proposed by the GoS may be made functional soon. This new policy will ensure effective water services delivery, though from the choice-based experiment conducted by this study, it was found that urban residents still prefer the existing water supply system. This indicates that the people are getting sufficient water at home, and it is also clear that they would not like to pay over and above what they are already doing with the subsidised tariff. However, with the upsurge of the population and urbanisation, the GoS does need to further expand the existing infrastructure for supplying sufficient and safe water to urban dwellers. The existing water tariff is very much underpriced and is now subject to a regressive pricing policy. Furthermore, under this present non-volumetric system, the wastage of water resources by users is very high. Therefore, the new/proposed volumetric water policy, though it is not purely a progressive tariff system, will definitely bring accountability on the part of water users and also be financially viable for the GoS.

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